Class: Algorithms & Data Structures Semester: Spring 2025

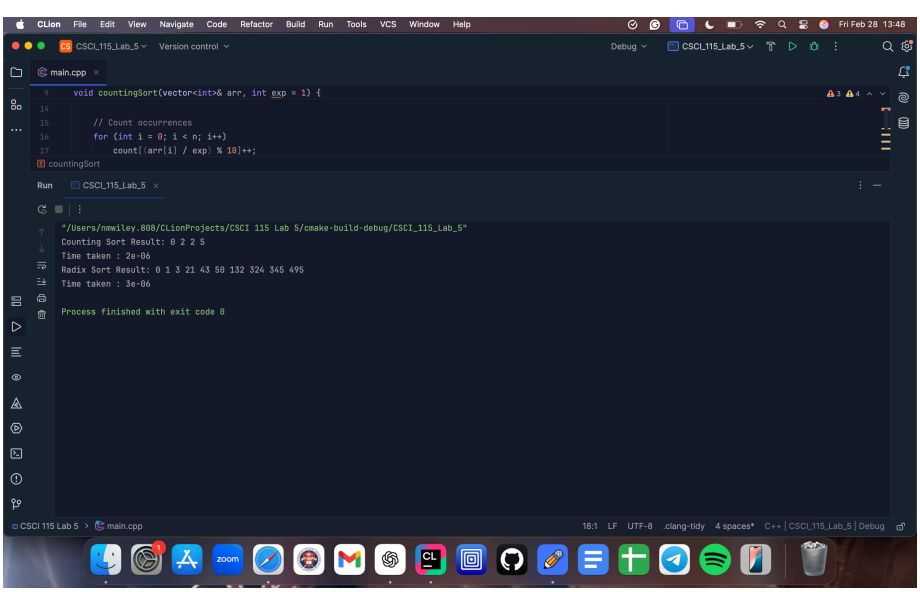
Noah Wiley
noahwiley@mail.fresnostate.edu
Lab Number - Section 09

**CSCI 115 Lab 5 Report; Counting & Radix Sort**

- Statement Of Objectives
  - The objective of this lab is to implement and analyze the performance of the Counting Sort and Radix Sort algorithms. The significance of this lab lies in understanding non-comparative sorting techniques and their efficiency in sorting numerical data. The major accomplishments include successfully coding the algorithms, executing on test data, and comparing their performance in terms of execution time. This report details the experimental procedure, analysis, encountered challenges, and conclusions drawn from this experiment.

- Experimental Procedure
  - The following steps were performed in this lab:
    - Implemented Counting Sort and Radix Sort in C++
    - Created test cases to evaluate both sorting algorithms
    - Measured execution time for each sorting method
    - Verified the correctness of the sorted outputs
    - Documented the results and analyzed their efficiency
  - This experiment was conducted using CLion IDE on macOS

- Analysis
  - The experimental results were obtained by running the sorting algorithms on predefined test data. The output and performance timings are as follows:



    - From the results, we can observe that Counting Sort performed efficiently on a smaller dataset, whereas Radix Sort effectively handled a larger dataset. The execution times indicate that both sorting techniques are highly efficient for their intended use cases.

- Encountered Problems
    - Some issues faced during the implementation:
        - Initially, the Counting Sort function had an indexing error due to incorrect use of counting array
        - Debugging memory allocation for vectors was necessary to ensure proper sorting

- Conclusions
    - Counting Sort is effective for small integer arrays, but has its limitations when handling a wider range of numbers
    - Radix Sort is suitable for sorting larger numerical datasets efficiently
    - Execution time measurement provided insight into the practical performance of each algorithm
    - The lab helped reinforce concepts related to non-comparative sorting algorithms and their applications

- References
    - Lab 5 Counting & Radix Sort Slides